

Mathematical Foundation of Support Vector Machines (SVM)

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What is an SVM?

Support Vector Machine (SVM) is a supervised learning algorithm primarily used for binary classification tasks.

Core goal: Find the *optimal* decision boundary (hyperplane) that maximally separates two classes in the feature space.

It is built on three mathematical pillars:

- **Maximum Margin Principle** — choose the hyperplane that is furthest from both classes, giving better generalisation.
- **Convex Optimisation** — the margin-maximisation problem is a convex quadratic programme, so a *global* optimum is always guaranteed.
- **Kernel Methods** — implicitly map data to high-dimensional feature spaces, enabling non-linear boundaries without ever computing the explicit mapping.

Why SVM? SVMs have strong theoretical guarantees (VC-dimension, structural risk minimisation) and work very well in high-dimensional, small-sample regimes such as text classification and bioinformatics.

Hyperplane and Margin — Geometric View

Given training data $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$ with $\mathbf{x}_i \in \mathbb{R}^d$, $y_i \in \{-1, +1\}$, a **hyperplane** in $\mathbb{R}^d$ is defined by

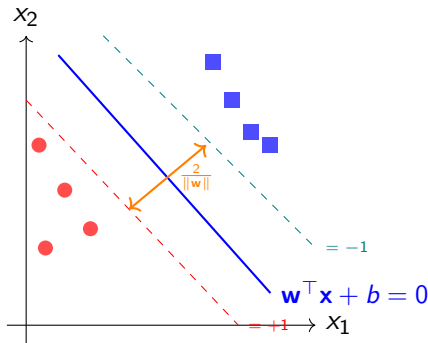
$$\mathbf{w}^\top \mathbf{x} + b = 0,$$

where $\mathbf{w} \in \mathbb{R}^d$ is the *normal vector* and $b \in \mathbb{R}$ is the *bias*.

Key quantities:

- **Positive margin plane:** $\mathbf{w}^\top \mathbf{x} + b = +1$
- **Negative margin plane:** $\mathbf{w}^\top \mathbf{x} + b = -1$
- **Margin width:** $\frac{2}{\|\mathbf{w}\|}$

The *geometric margin* is the perpendicular distance from the hyperplane to the nearest data points.



Hard-Margin SVM: Primal Optimisation

When data are **linearly separable**, we seek

$$\min_{\mathbf{w}, b} \quad \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{subject to} \quad y_i (\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, \quad \forall i.$$

Why $\frac{1}{2} \|\mathbf{w}\|^2$?

- Maximising the margin $\frac{2}{\|\mathbf{w}\|}$ is equivalent to *minimising* $\|\mathbf{w}\|^2$.
- The factor $\frac{1}{2}$ is a convenience that simplifies the gradient: $\nabla_{\mathbf{w}} \frac{1}{2} \|\mathbf{w}\|^2 = \mathbf{w}$.

Why $y_i (\mathbf{w}^\top \mathbf{x}_i + b) \geq 1$?

- The canonical constraint forces the closest points (*support vectors*) to lie *on* the margin planes ± 1 .
- Points correctly classified far from the boundary satisfy the constraint with *strict* inequality.

This is a convex QP: The objective is quadratic and the constraints are linear, so any local minimum is a global minimum.

Lagrangian and Dual Formulation

Introduce Lagrange multipliers $\alpha_i \geq 0$ for each constraint:

$$\mathcal{L}(\mathbf{w}, b, \alpha) = \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^n \alpha_i [y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1].$$

Setting partial derivatives to zero gives the KKT stationarity conditions:

$$\mathbf{w} = \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i, \quad \sum_{i=1}^n \alpha_i y_i = 0.$$

Substituting back yields the **Wolfe dual**:

$$\max_{\alpha} \quad \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j \quad \text{s.t.} \quad \sum_{i=1}^n \alpha_i y_i = 0, \quad \alpha_i \geq 0.$$

Why the dual matters:

- Data appear only via *dot products* $\mathbf{x}_i^\top \mathbf{x}_j$ — these can be replaced by a **kernel function** $K(\mathbf{x}_i, \mathbf{x}_j)$.

Support Vectors and KKT Conditions

KKT complementary slackness requires

$$\alpha_i [y_i(\mathbf{w}^\top \mathbf{x}_i + b) - 1] = 0, \quad \forall i.$$

This means:

- If $\alpha_i > 0$: the constraint is *active*, i.e. $y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1$. These points are called **support vectors** — they lie exactly on the margin planes.
- If $\alpha_i = 0$: the point is correctly classified with a *strict* margin and has **no influence** on $\mathbf{w}$ or b .

Sparsity: In practice only a small fraction of training points are support vectors. This makes SVM memory-efficient at test time:

$$f(\mathbf{x}) = \text{sign} \left(\sum_{i: \alpha_i > 0} \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b \right).$$

Recovering b : Use any support vector $\mathbf{x}_s$:

Soft-Margin SVM: Handling Non-Separable Data

Real data are rarely linearly separable. We introduce **slack variables** $\xi_i \geq 0$ to allow violations:

$$\min_{\mathbf{w}, b, \xi} \quad \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

Interpretation of ξ_i :

- $\xi_i = 0$: point correctly classified, outside margin.
- $0 < \xi_i \leq 1$: point inside margin but correct side.
- $\xi_i > 1$: point *misclassified*.

Role of the regularisation parameter $C > 0$:

- **Large** C : penalises violations heavily $\Rightarrow$ narrow margin, risk of overfitting.
- **Small** C : tolerates violations $\Rightarrow$ wide margin, risk of underfitting.

The dual has the same form as before but with bounded multipliers: $0 \leq \alpha_i \leq C$.

Kernel Trick: Non-Linear Boundaries

Many problems are not linearly separable in $\mathbb{R}^d$ but become separable in a higher-dimensional feature space.

Feature map: $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^D$, $D \gg d$.

Replace every dot product $\mathbf{x}_i^\top \mathbf{x}_j$ in the dual with a **kernel function**:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^\top \phi(\mathbf{x}_j).$$

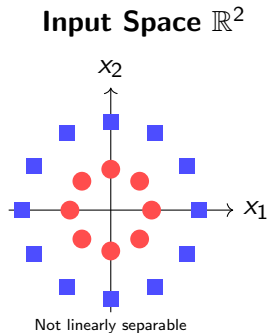
Key insight (Kernel Trick): We never compute ϕ explicitly. We only need to evaluate K , which is often cheap even when $D = \infty$ (e.g. RBF kernel).

Mercer's theorem guarantees that any symmetric positive semi-definite function K corresponds to some inner product in some feature space — it is a valid kernel.

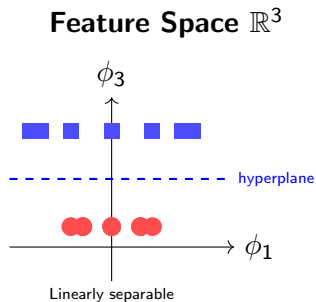
The decision function becomes:

$$f(\mathbf{x}) = \text{sign} \left(\sum_{i=1}^n \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b \right).$$

Kernel Transformation — Illustration



ϕ



The RBF kernel achieves this implicitly with $D = \infty$.

Common Kernel Functions

Kernel	Formula	Use case
Linear	$K(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\top \mathbf{x}'$	Linearly separable data
Polynomial	$K(\mathbf{x}, \mathbf{x}') = (\mathbf{x}^\top \mathbf{x}' + c)^d$	Interaction features
RBF (Gaussian)	$K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma \ \mathbf{x} - \mathbf{x}'\ ^2)$	General-purpose
Sigmoid	$K(\mathbf{x}, \mathbf{x}') = \tanh(\kappa \mathbf{x}^\top \mathbf{x}' + c)$	Neural-net analogy

Notes on the RBF kernel:

- $\gamma > 0$ controls the *width* of the Gaussian bump.
- Large γ : narrow influence region, complex boundary (overfitting risk).
- Small γ : smooth, wide influence (underfitting risk).
- Corresponds to an infinite-dimensional feature space.

Key Takeaways

- 1 **Maximum-margin hyperplane** — SVM finds the unique separating hyperplane with the largest geometric margin, giving better generalisation.
- 2 **Convex QP + Lagrangian duality** — the primal problem is converted to a dual that depends only on dot products; strong duality holds (Slater's condition).
- 3 **Support vectors** — only the points on or inside the margin (with $\alpha_i > 0$) define the solution; all others are irrelevant.
- 4 **Soft-margin / regularisation** — the parameter C trades off margin width vs. training error.
- 5 **Kernel trick** — replacing dot products with K enables non-linear boundaries without explicit feature maps.
- 6 **Applications** — text classification, image recognition, bioinformatics, and SVR (regression).